
BASIC STATISTICS-2

Problem Statement : Hospital Patient Data Analysis

Context:

A hospital maintains patient records including admission details, department, diagnosis, doctor, and bill amount. You have two datasets: one with patient info and another with billing details. Some patients have blank bill amounts, and there are multiple rows for the same patient due to follow-ups.

Tasks:

1. Load the patient dataset and show summary with `info()`.
2. Select only the columns relevant for billing: ['PatientID', 'Department', 'Doctor', 'BillAmount'].
3. Drop administrative columns like ['ReceptionistID', 'CheckInTime'].
4. Use `groupby` to find **total bill amount per department**.
5. Remove **duplicate** patient records based on PatientID.
6. Fill missing BillAmount values with the **mean** bill amount.
7. Merge the billing dataset with patient dataset on PatientID.
8. Concatenate an additional DataFrame that contains new patients for the current week (row-wise).
9. Concatenate new billing category columns like ['InsuranceCovered', 'FinalAmount'] (column-wise).

Expected Outcome:

- Final cleaned dataset with accurate billing info.
- All missing values handled, merged dataset across PatientID.
- Ability to perform further analytics on department-wise revenue or doctor performance.